Erica Bertolozzi

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Frontend-focused full-stack engineer specializing in building intuitive, high-impact user interfaces backed by robust React and Python systems. Experienced in guiding junior engineers, shaping frontend architecture, and collaborating across product, QA, and operations teams. Known for turning stakeholder needs into clean, scalable solutions with strong emphasis on usability, clarity, and maintainability.

Work Experience

Deloitte Consulting - New York, NY

Software Engineer II *Jun. 2025 - Present*

- Led the front-end development of seven generative AI-powered risk analysis tools, driving over 99% reduction in time required to generate cybersecurity control artifacts (from weeks to 10 minutes), demoed to agency stakeholders;
- Architected factory-based multi-agent system supporting 6 specialized AI security analysts, reducing agent instantiation code by 70% and enabling dynamic model switching;
- Engineered cloud security agent detecting 15+ threat types using MCP-based Athena queries, processing CloudTrail logs with <2s query turnaround;
- Integrated AWS Bedrock foundation models via custom provider abstraction, enabling cost-optimized model selection;
- Designed and implemented 16 user interfaces end-to-end using Figma, collaborating with QA teams for rigorous testing while ensuring compliance with Deloitte branding standards and WCAG accessibility guidelines;
- Engineered 70 API integrations with Python/Flask and PostgreSQL deployed in Docker, enabling seamless data flow and supporting high-velocity feature development across all analysis tools;
- Developed full-stack React/Python resume automation tool with MongoDB data layer, Microsoft SSO, Azure QA deployment pipeline and CI/CD integration, accelerating production go-live by 3 weeks.

Software Engineer I *Sept. 2022 – Jun. 2025*

- Enhanced security and operational efficiency by architecting and implementing role-based access control across 3 permission levels, securing 5 critical workflows and 20+ API endpoints for order placement and inventory management, ensuring manufacturing clients operated within a guarded and auditable permission framework;
- Created and optimized 3 real-time factory simulation dashboards in React, visualizing 12+ KPIs and 15+ interactive charts to help clients identify bottlenecks and address inefficiencies. Pre-launch testing showed 20–30% faster issue detection and projected 10–15% reductions in plant downtime once deployed;
- Developed and deployed an interactive changeover time widget tracking 8 operational parameters, equipping users with actionable insights that drove continuous process improvement and supported data-driven decision-making on the factory floor.

Business Technology Analyst *Sept. 2021 – Sept. 2022*

- Directed end-to-end UAT for a client-facing loan portal, executing 550 test cases and identifying 800+ defects in the third-party vendor platform. Partnered closely with the vendor to validate fixes and ensure full alignment with business requirements ahead of launch.

Business Technology Summer Scholar *July 2020*

- Completed a virtual consulting program developing client engagement and problem-solving skills through a simulated project optimizing restaurant operations during the COVID-19 pandemic.

General Dynamics Mission Systems - Pittsfield, MA

Software Engineer Intern *May 2019 – Jul. 2019*

- Developed a high-performance simulation tool using C++ socket programming and multithreading, modeling critical system interactions for an early-stage defense project.
- Contributed to mission-critical software development under a U.S. Department of Defense security clearance, handling confidential information with precision and integrity.

Education

Rensselaer Polytechnic Institute (RPI), Bachelor's Degree in Computer Science May 2021

Skills and Certifications

Languages: React, TypeScript, Next.js, Python, Java, .NET/C#, HTML

APIs and Databases: RESTful APIs, Flask, PostgreSQL, MongoDB

Technologies: git, Docker, Figma, Firebase, Claude Code, Google ADK, AWS Bedrock

Certifications: Google Cloud GenAI Leader, Google Cloud Digital Leader

Interests: Yoga, Weightlifting, Video Games, Concerts, Puzzles,

Leadership: Young Adult Group Leader for Sts. Peter and Paul Catholic Church